

Computational Fluid Dynamics

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We want to simulate the non-isothermal flow of a non-Newtonian fluid through a two-dimensional rectangular duct $\Omega = (0, 4) \times (0, 1)$. The problem is governed by the steady non-Newtonian incompressible Navier-Stokes equations coupled with the temperature equation

$$\left\{ \begin{array}{ll} -\operatorname{div}(2\mu_{\text{eff}}\mathbf{D}(\mathbf{u})) + \rho(\mathbf{u} \cdot \nabla)\mathbf{u} + \nabla p = \mathbf{0} & \text{in } \Omega \\ \operatorname{div} \mathbf{u} = 0 & \text{in } \Omega, \\ -k\Delta T + \rho c_p \mathbf{u} \cdot \nabla T = 2\mu_{\text{eff}}\mathbf{D}(\mathbf{u}) : \mathbf{D}(\mathbf{u}) & \text{in } \Omega, \\ \mathbf{u} = (10y(1-y), 0)^T, \quad T = T_0 & \text{on } \Gamma_1 = \{(x, y) \in \partial\Omega : x = 0\}, \\ (-p\mathbf{I} + 2\mu_{\text{eff}}\mathbf{D})\mathbf{n} = \mathbf{0}, \quad \frac{\partial T}{\partial \mathbf{n}} = 0 & \text{on } \Gamma_2 = \{(x, y) \in \partial\Omega : x = 4\}, \\ \mathbf{u} = \mathbf{0}, \quad \frac{\partial T}{\partial \mathbf{n}} = 0 & \text{on } \Gamma_{3,4} = \{(x, y) \in \partial\Omega : y = 0 \text{ or } y = 1\}, \end{array} \right. \quad (1)$$

where $\mathbf{D}(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla^T \mathbf{u})$ is the rate of deformation tensor, ρ the density, k the thermal conductivity and c_p the heat capacity. The effective viscosity $\mu_{\text{eff}} = \mu_{\text{eff}}(\dot{\gamma}, T)$ depends on the shear rate $\dot{\gamma} = \sqrt{2\mathbf{D}(\mathbf{u}) : \mathbf{D}(\mathbf{u})}$ through the so-called power-law and on the temperature T through the so-called Arrhenius model, namely

$$\mu_{\text{eff}}(\dot{\gamma}, T) = \mu_0 \exp(T_0/T) \dot{\gamma}^{n-1}$$

for given constants μ_0 , T_0 and n .

- a) Introduce a suitable linearization of problem (1) and the weak formulation of the linear problem to be solved at each nonlinear iteration.
- b) Starting from the template `template070726.ipynb`, write a `Firedrake` notebook for the solution of problem (1) using $\mathbb{P}_2$ finite elements for velocity, $\mathbb{P}_1$ for pressure and $\mathbb{P}_1$ for temperature. As an initial guess, consider the Stokes solution with $\mu = \mu_0$ and uniform temperature $T = T_0$. Define and use a suitable convergence criterion for the nonlinear iterations.
- c) Solve the nonlinear problem for $\rho = 1$, $c_p = 1$, $\mu_0 = 1$, $T_0 = 300$, $k = 1$ and three different values of $n = 1, 0.6, 0.2$, reporting the number of nonlinear iterations required to reach convergence for each value of n .
- d) Postprocess the final result for the three cases $n = 1, 0.6, 0.2$, saving the velocity, pressure, and temperature plots as `SURNAME_U.n.png`, `SURNAME_P.n.png`, and `SURNAME_T.n.png`, respectively.
- e) Compute and report the x component of the force exerted by the flow on the duct walls Γ_3 and Γ_4 for the different values of n .
- f) Comment the results.